La Scuola d'Italia / Dr. Huson / IB Math (c)
14 March 2026

Student ID:

7.11 PreQuiz: Logarithmic and Exponential Functions (calculator)

1. [Maximum mark: 6]

   Noah invests \$3200 in an account that pays interest of 7.2 % per annum, compounded quarterly. No further deposits or withdrawals are made. Let $V(t)$ be the value of the account after t years.

   (a) Write down a formula for $V(t)$ in the form $V(t) = a(b)^{4t}$. [2]

   (b) Write the same model in the equivalent continuous form $V(t) = 3200e^{kt}$, giving the exact value of k. [2]

   (c) Find the smallest number of full years required for the value of the account to first exceed \$5000. [2]

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2. [Maximum mark: 6]

The number of users of a new app is modelled by

$$U(t) = 9000e^{kt}, \qquad k > 0$$

where t is the number of years after launch. After 3 years there are 18 500 users.

(a) Find the value of k. [2]

(b) Find the number of users after 7 years, correct to the nearest whole number. [2]

(c) Find the time, in years, when the number of users first reaches 50 000. Give your answer correct to two decimal places. [2]